EDUCATIONAL ORGANISATION MANAGEMENT SYSTEM

Using ServiceNow Platform

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1. INTRODUCTION

1.1 Project Overview

The Educational Organisation Management System is a cloud-based application developed on the ServiceNow platform to automate and manage core administrative activities within an educational institution. The system enables structured management of student records, teacher records, course enrollment, and academic progress tracking.

Traditionally, many educational institutions rely on manual record-keeping systems such as paper files or Excel spreadsheets. These systems lead to inefficiencies, duplication of data, and difficulty in tracking student progress over time. To overcome these challenges, this project implements a centralized, scalable, and structured solution using ServiceNow's low-code development environment.

The system is designed using custom tables, reference fields, modules, and reports to ensure accurate data storage and efficient retrieval. It demonstrates the practical application of enterprise workflow automation in the education sector.

1.2 Purpose

The primary purpose of this project is to:

- Automate administrative processes in educational institutions
- Centralize student, teacher, and enrollment data
- Improve data accuracy and consistency
- Provide structured course enrollment management
- Enable better reporting and academic tracking
- Demonstrate understanding of ServiceNow application development

2. IDEATION PHASE

2.1 Problem Statement

Educational institutions face several operational challenges:

- Student records stored across multiple spreadsheets
- Manual admission processes
- Lack of centralized teacher and course data

- Difficulty in linking students to courses
- Time-consuming data retrieval
- High risk of duplication and inconsistency

There is a need for a centralized, automated system to handle educational data management efficiently and securely.

2.2 Empathy Map Canvas

Target User: Administrative Staff

Thinks: Managing student and teacher data manually is inefficient. There must be a better digital system.

Feels: Overwhelmed by large volumes of records. Frustrated by repeated manual tasks.

Sees: Multiple spreadsheets. Paper admission forms.

Does: Manually updates records. Cross-checks data frequently.

Pain Points: Data inconsistency. Slow report generation. Duplicate entries.

Goals: A centralized, simple, and automated system.

2.3 Brainstorming

During brainstorming, several approaches were considered: Spreadsheet automation, Custom web application using coding frameworks, Cloud-based workflow automation platform.

The final decision was to develop the system on ServiceNow due to built-in database management, low-code development, cloud deployment, and enterprise-grade reliability.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Admin logs into system → Creates teacher records → Creates student records → Assigns courses → Enrolls students into courses → Updates student academic progress → Generates reports.

3.2 Solution Requirements

Functional Requirements: Create Student Records, Create Teacher Records, Manage Course Details, Enroll Students into Courses, Track Student Progress, Generate Reports.

Non-Functional Requirements: Usability (Easy-to-use forms), Reliability, Performance (Fast retrieval), Availability (Cloud-based), Scalability.

3.3 Data Flow Diagram (Textual Description)

Admin → ServiceNow Form → Custom Tables → Reports

Flow Explanation: Admin enters student/teacher data. Data stored in respective tables. Course enrollment links student and teacher records. Reports retrieve data from tables.

3.4 Technology Stack

Layer	Technology Used
Platform	ServiceNow
Development	Low-code configuration
Database	ServiceNow Tables
Reporting	ServiceNow Reports
Deployment	ServiceNow Cloud (PDI)

4. PROJECT DESIGN

4.1 Problem – Solution Fit

The solution directly addresses administrative inefficiencies by centralizing data, automating record storage, and enabling real-time reporting.

4.2 Proposed Solution

The proposed system consists of Student Table, Teacher Table, Course Table, Enrollment Table, and automated Reports for data visualization.

4.3 Solution Architecture

Architecture Type: 3-Tier Architecture

- Presentation Layer: ServiceNow Forms
- Application Layer: Business Logic, Table relationships

• Data Layer: Custom Tables, Reference fields

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Development followed Agile methodology: Sprint 1 (Table creation), Sprint 2 (Enrollment and reporting). Team velocity: 16 story points per sprint.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

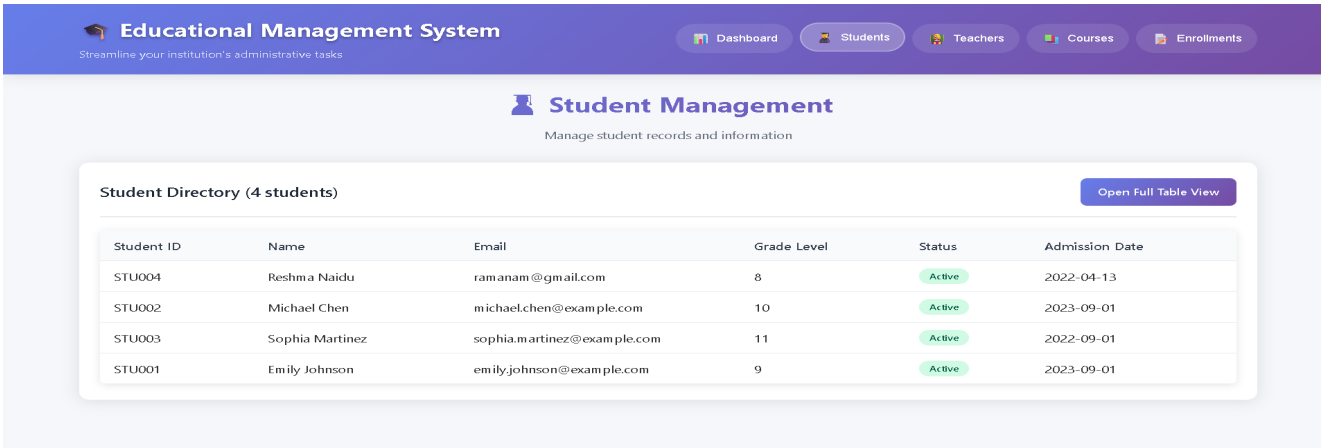
System tested for record creation accuracy, enrollment linkage, data retrieval speed, and report accuracy. All test cases passed successfully.

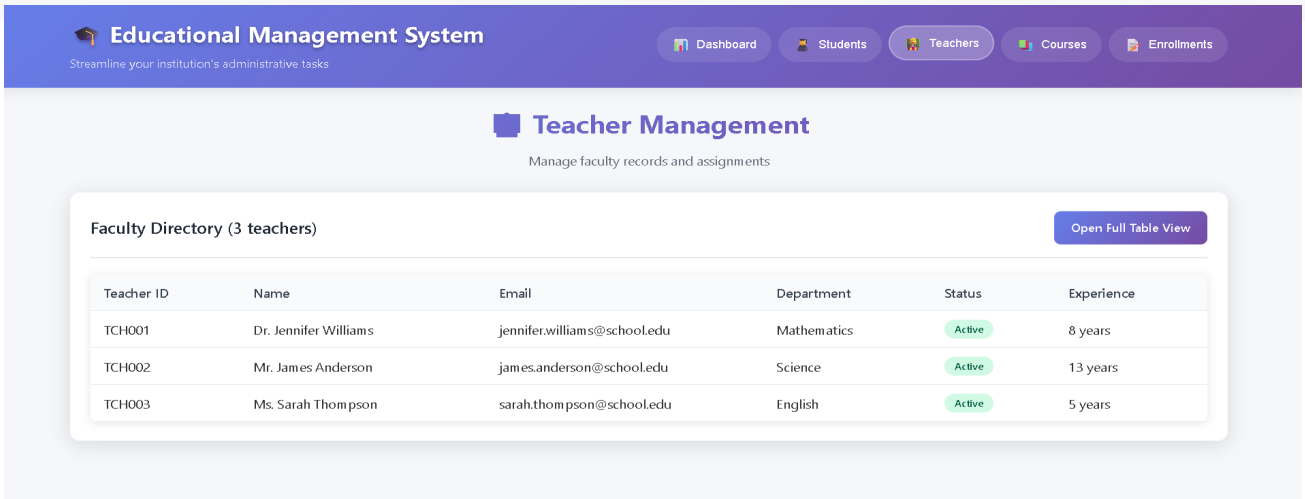
7. RESULTS

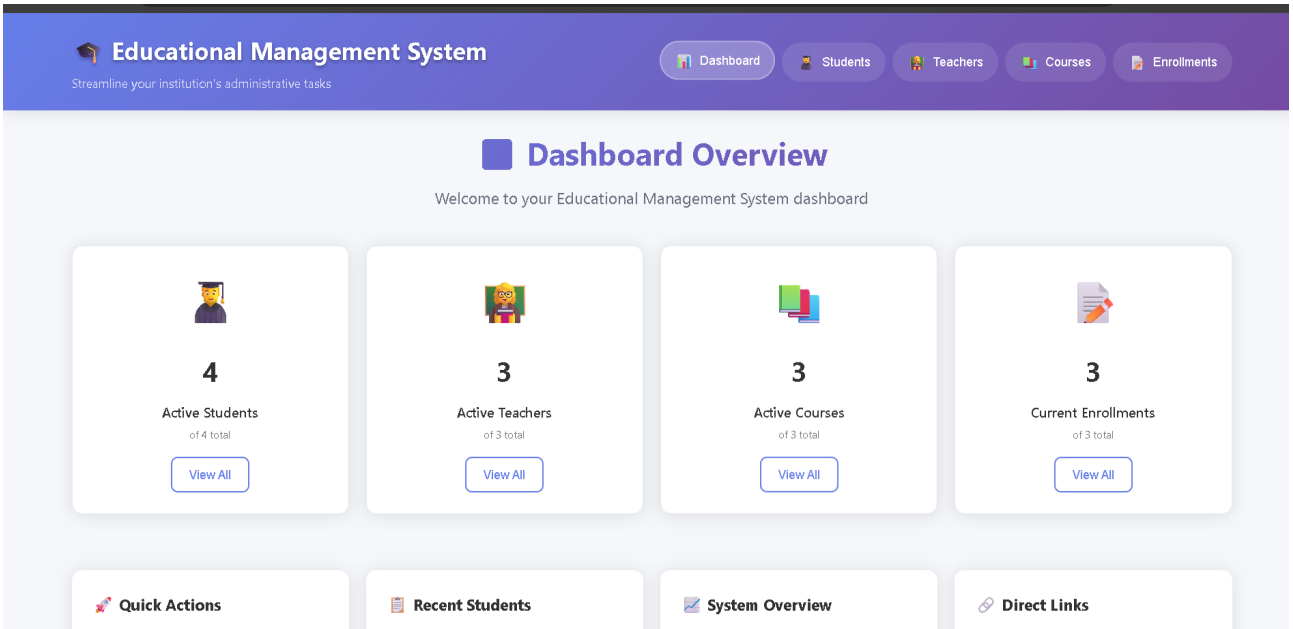
7.1 Output Screenshots

Student Record Creation Screen:

Teacher Record Screen:







Report Output Screen:

8. ADVANTAGES & DISADVANTAGES

Advantages: Centralized management, reduced manual work, scalable solution, cloud-based.

Disadvantages: Dependent on ServiceNow platform, requires internet connectivity.

9. CONCLUSION

The Educational Organisation Management System successfully demonstrates the use of ServiceNow as an enterprise automation platform beyond IT service management. The system improves efficiency, reduces manual effort, and ensures structured data management.

10. FUTURE SCOPE

Attendance management, Fee management, email notifications, and Advanced analytics dashboards.

11. APPENDIX

Source Code

Not applicable (Low-code ServiceNow platform) but custom source can be used for ui related or even scripts like:

```
function onChange(control, oldValue, newValue, isLoading, isTemplate) {
   if (isLoading || newValue === '') {
```

```
        return;
    }

    //Type appropriate comment here, and begin script below
    var a = g_form.getReference('u_admission_number');
    g_form.setValue('u_admin_date',a.u_admin_date);
    g_form.setValue('u_grade',a.u_grade);
    g_form.setValue('u_student_name',a.u_student_name);
    g_form.setValue('u_father_name',a.u_father_name);
    g_form.setValue('u_mother_name',a.u_mother_name);
    g_form.setValue('u_father_cell',a.u_father_cell);
    g_form.setValue('u_mother_cell',a.u_mother_cell);

    g_form.setDisabled('u_admin_date',true);
    g_form.setDisabled('u_grade',true);
    g_form.setDisabled('u_student_name',true);
    g_form.setDisabled('u_father_name',true);
    g_form.setDisabled('u_mother_name',true);
    g_form.setDisabled('u_father_cell',true);
    g_form.setDisabled('u_mother_cell',true);
}
```

Manual data entry / Dataset Link: CSV manual imports.

GitHub & Project Demo Link: https://github.com/Sujana2004/Education-Management-System-ServiceNow

ServiceNow Developer Instance (PDI): https://dev285953.service-now.com/x_1917234_educatio_dashboard.do